Class 20 Learning Activity

Matlab Work-Up

Before summarizing the assignment, the file, `act_20.m`, was created for performing the calculations. Within it a function of the same name was declared. Each of the blocks of code listed below are added to this file to complete the function for performing the calculations.

```
function act_20()
% Calculations for the Class 20 Learning Activity from REB, The Book
```

Assignment Summary

Reactions

$$A + B \rightarrow D \tag{1}$$

$$2B \rightarrow U \tag{2}$$

Rate Expressions

$$r_1 = k_1 C_A C_B \tag{3}$$

$$r_2 = k_2 C_B^2 \tag{4}$$

Reactor: Adiabatic, liquid-phase SBSTR

Given Constants: $P = 1$ atm, $V_0 = 550$ gal, $T_0 = 100$ °F, $C_{A,0} = 0.015$ lbmol gal^{-1}, $V_B = 2000$ gal, $T_{in} = 100$ °F, $C_{B,in} = 0.004$ lbmol gal^{-1}, $f_{B,min} = 0.9$, $t_{turn} = 30$ min, $k_{0,1} = 4.193$ x 10^{11} gal lbmol^{-1} h^{-1}, $E_1 = 32{,}400$ BTU lbmol^{-1}, $\Delta H_1 = -16{,}190$ BTU lbmol^{-1}, $k_{0,2} = 6.087$ x 10^{14} gal lbmol^{-1} h^{-1}, $E_2 = 36{,}900$ BTU lbmol^{-1}, $\Delta H_2 = -14{,}030$ BTU lbmol^{-1}, and $\breve{C}_p = 7.2$ BTU gal^{-1} °R^{-1}.

The given and known constants and the basis are converted to consistent units and the given equations are converted to anonymous functions and all are addded to `act_20`.

```
% global constants available to all functions
% given
P = 1.0; % atm
V0 = 550.; % gal
T0 = 100 + 459.7; % °R
CA0 = 0.015; % lbmol/gal
VB = 2000.; % gal
Tin = 100 + 459.7; % °R
CBin = 0.004; % lbmol/gal
fB_min = 0.9;
tTurn = 0.5; % h
k01 = 4.193e+11; % gal/lbmol/h
E1 = 32400; % BTU/lbmol
dH1 = -16190; % BTU/lbmol
k02 = 6.087e+14; % gal/lbmol/h
E2 = 36900; % BTU/lbmol
dH2 = -14030; % BTU/lbmol
Cp = 7.2; % BTU/gal/°R
% known
R = 1.987; % BTU/lbmol/°R
Rpv = 5.4584; % gal-atm/lbmol/°R
% calculated
nA0 = CA0*V0;
```

Task: Optimization

- Parameter: $\dot{V}$
- Optimization Target: $r_{D,net}$
- Items of Interest: $\dot{V}_{opt}$ and $r_{D,net,max}$

Preliminary Analysis

Ignoring temperature effects, D is only produced in reaction (1). The rate of reaction (1) depends upon the concentrations of both A and B. A only reacts in reaction (1), so maintaining the concentration of A as high as possible will favor production of D. Be reacts in both reactions, and its rate of consumption in the undesired reaction depends more strongly on its concentration that its rate of consumption in the desired reaction. Maintaining the concentration of B as low as possible will disfavor its conversion to the undesired product. Thus, by charging the reactor with A (so its concentration is a high as possible) and slowly adding B (so its concentration is as low as possible) should favor the production of the desired product, D.

SBSTR Model Function

SBSTR Design Equations

$$\frac{dn_A}{dt} = -r_1 V \tag{5}$$

$$\frac{dn_B}{dt} = \dot{n}_{B,in} - (r_1 + 2r_2)\, V \tag{6}$$

$$\frac{dn_D}{dt} = r_1 V \tag{7}$$

$$\frac{dn_U}{dt} = r_2 V \tag{8}$$

$$\frac{dT}{dt} = \frac{-\dot{V}_{in}\breve{C}_p\,(T - T_{in}) - r_1 V \Delta H_1 - r_2 V \Delta H_2 + P\dot{V}_{in}}{V\breve{C}_p} \tag{9}$$

$$\frac{dV}{dt} = \dot{V}_{in} \tag{10}$$

SBSTR Model Variables: $\underline{t}$, $\underline{n}_A$, $\underline{n}_B$, $\underline{n}_D$, $\underline{n}_U$, $\underline{T}$, and $\underline{V}$

Additional Computable Unknowns: r_1, r_2, k_1, k_2, C_A, C_B, and $\dot{n}_{B,in}$

$$k_i = k_{0,i} \exp\left(\frac{-E_i}{RT}\right) \qquad i = 1 \text{ and } 2 \tag{11}$$

$$C_i = \frac{n_i}{V} \qquad i = A \text{ and } B \tag{12}$$

$$\dot{n}_{B,in} = C_{B,in}\dot{V}_{in} \tag{13}$$

Uncomputable Unknowns (Received as Arguments): $\dot{V}_{in}$

$$\underline{\dot{V}}_{in} = [20, \cdots, 40] \text{ gal h}^{-1} \tag{14}$$

(The derivatives function needs $\dot{V}_{in}$, so the SBSTR model function must make it available as a global variable.)

IVODE Solver Requirements

1.1 Stage 1 initial values

$$t_{iv} = 0 \tag{15}$$

$$n_{A,iv} = n_{A,0} = CA, 0V_0 \tag{16}$$

$$n_{i,iv} = 0 \qquad i = B, D, \text{ and } U \tag{17}$$

$$T_{iv} = T_0 \tag{18}$$

$$V_{iv} = V_0 \tag{19}$$

2.1 Stage 1 stopping criterion

$$V = V_f = V_0 + V_B \tag{20}$$

1.2 (If $\frac{C_{B,in}V - n_{B,f,1}}{C_{B,in}V} < f_{B,min}$) Stage 2 initial values

$$t_{iv} = t_{1,f} \tag{21}$$

$$n_{i,iv} = n_{i,f,1} \qquad i = A, B, D, \text{ and } U \tag{22}$$

$$T_{iv} = T_{1,f} \tag{23}$$

$$V_{iv} = V_{1,f} \tag{24}$$

2.2 (If $\frac{C_{B,in}V - n_{B,f,1}}{C_{B,in}V} < f_{B,min}$) Stage 2 stopping criterion

$$n_B = V_B C_{B,in} \left(1 - f_{B,min}\right) \tag{25}$$

3. SBSTR derivatives function that

a. receives the independent and dependent variables (t, n_A, n_B, n_D, n_U, T, and V) at the start of an integration step

b. can access the current value of $\dot{V}_{in}$

c. calculates the additional unknowns using equations (3), (4), (11), (12), and (13)

d. evaluates and returns the SBSTR design equation derivatives using equations (5) through (10).

To facilitate making $\dot{V}_{in}$ available to the SBSTR derivatives function, the global variable, `g_Vdot` is declared and set to `nan` initially. Then the SBSTR model and derivatives functions are added to `act_20`.

```
% global variable for the current volumetric flow rate
g_Vdot = nan;

% SBSTR reactor function
function [t, nA, nB, nD, nU, T, V] = sbstr_model_variables(Vdot)
    % set the volumetric flow rate for stage 1
    g_Vdot = Vdot;

    % set the initial values for the first stage
    ind_0 = 0;
    dep_0 = [nA0, 0, 0, 0, T0, V0];

    % set the stopping criterion
    stopVar = 6;
    stopVal = V0 + VB;

    % solve the design equations for stage 1
    odes_are_stiff = false;
    [t, dep, success, message] = solve_ivodes(ind_0, dep_0, stopVar...
        , stopVal, @sbstr_derivatives, odes_are_stiff);

    % extract the individual dependent variables
    nA = dep(:,1);
    nB = dep(:,2);
    nD = dep(:,3);
    nU = dep(:,4);
    T = dep(:,5);
    V = dep(:,6);

    % check for solver issues
```

```
        if ~success
            disp('')
            disp(["SBSTR Model stage 1 issue: ", message])
            disp('Press any key to continue or CTRL-C to exit')
            pause
        end

        % calculate the conversion of B during stage 1
        fB1 = 100*(VB*CBin - nB(end))/(VB*CBin);

        % if the conversion is below 90% continue with stage 2
        if fB1 < 90

            % set the volumetric flow rate for stage 2
            g_Vdot = 0;

            % set the initial values equal to the final values from stage 1
            ind_0 = t(end);
            dep_0 = [nA(end), nB(end), nD(end), nU(end), T(end), V(end)];

            % set the stopping criterion for stage 2
            stopVar = 2;
            stopVal = VB*CBin*(1 - fB_min);

            % solve the design equations for stage 2
            odes_are_stiff = false;
            [t2, dep, success, message] = solve_ivodes(ind_0, dep_0, stopVar...
                , stopVal, @sbstr_derivatives, odes_are_stiff);

            % combine the results for the two stages
            t = [t; t2];
            nA = [nA; dep(:,1)];
            nB = [nB; dep(:,2)];
            nD = [nD; dep(:,3)];
            nU = [nU; dep(:,4)];
            T = [T; dep(:,5)];
            V = [V; dep(:,6)];

            % check for solver issues
            if ~success
                disp('')
                disp(["SBSTR Model stage 2 issue: ", message])
```

```
            disp('Press any key to continue or CTRL-C to exit')
            pause
        end
    end
end

% SBSTR derivatives function
function ddt = sbstr_derivatives(~, dep)
    % Extract the dependent variables
    nA = dep(1);
    nB = dep(2);
    T = dep(5);
    V = dep(6);

    % calculate the additional unknowns
    k1 = k01 * exp(-E1/(R*T));
    k2 = k02 * exp(-E2/(R*T));
    CA = nA / V;
    CB = nB / V;
    r1 = k1 * CA * CB;
    r2 = k2 * CB^2;
    nB_in = g_Vdot * CBin;

    % Calculate the derivatives
    dnAdt = -V*r1;
    dnBdt = nB_in - V*(r1 + 2*r2);
    dnDdt = V*r1;
    dnUdt = V*r2;
    dTdt = (-g_Vdot*Cp*(T-Tin)-V*r1*dH1 -V*r2*dH2 + P*g_Vdot*R/Rpv)/V/Cp;
    dVdt = g_Vdot;

    % return the derivatives as a column vector
    ddt = [dnAdt; dnBdt; dnDdt; dnUdt; dTdt; dVdt];
end
```

Items of Interest Function

1. define the range of values of $\dot{V}_{in}$, equation (14)
2. loop through that range of values
    a. calling the SBSTR model function for each one to get $\underline{t}$ and $\underline{n}_D$ and

b. calculating and saving the corresponding net rate

$$r_{D,net} = \left. \frac{n_{D,f,2}}{t_{f,2} + t_{turn}} \right|_{\dot{V}_{in}} \tag{26}$$

3. determine the maximum net rate and the corresponding volumetric flow rate

$$r_{D,net,max} = \max\left(r_{D,net}\right) \tag{27}$$

$$\dot{V}_{opt} = \arg\max_{\dot{V}_{in}}\left(r_{D,net}\right) \tag{28}$$

4. show, and save the results.

To complete `act_20`, the items of interest function was added along with a line of code to call it and perform the calculations.

```
% items of interest function
function items_of_interest()
    % set a range of values for the volumetric flow rate
    %Vdot_range = linspace(20, 40, 100)
    Vdot_range = linspace(5, 20, 100); % narrow the range

    % allocate storage for the corresponding net rates
    rNet = ones(100,1)*nan;

    % for each value of the volumetric flow rate
    for i = 1:100
        % solve the design equations
        [t, ~, ~, nD, ~, ~, ~] = sbstr_model_variables(Vdot_range(i));

        % calculate and save the corresponding net rate
        rNet(i) = nD(end)/(t(end) + tTurn);
    end

    % find the index of the maximum net rate and corresponding Vdot
    [max_rNet, iMax] = max(rNet);
    Vdot_opt = Vdot_range(iMax);

    % tabulate, show and save the results
    item = ["Optimum Feed Rate"; "Net Rate"];
```

```
        value = [Vdot_opt; max_rNet];
        units = ["gal/h"; "lbmol/h"];
        resultsTable = table(item, value, units);
        disp('')
        disp(resultsTable)
        disp('')
        writetable(resultsTable, 'act_20_results.csv')

        % plot the net rate vs. the feed rate
        figure
        plot(Vdot_range, rNet, 'LineWidth', 2)
        xlabel('Feed Rate (gal/h)', 'FontSize', 18)
        ylabel('Net Rate (lbmol/h)','FontSize', 18)
        set(gca,'FontSize', 18)
        saveas(gcf,'act_20_rNet_vs_VFR.png')
    end

    % perform the calculations
    items_of_interest()
end
```

Results

The calculations were performed by calling the function, `act_20`. The range of values of $\dot{V}_{in}$ was made narrower so the maximum could be more accurately determined. The results are shown below and in Figure 1.

```
>> act_20
           item              value         units
    ___________________    __________    _________

    "Optimum Feed Rate"        11.061    "gal/h"
    "Net Rate"             0.00075106    "lbmol/h"
```

Discussion

The instantaneous selectivity is the rate of production of the desired product to that of the undesired product. For this system, the selectivity for D over U is given in equation (29). It clearly shows that high concentration of A and low concentration of B give the

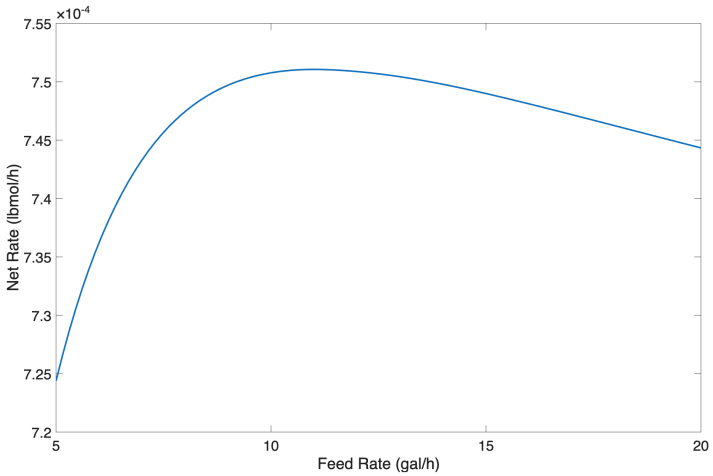


Figure 1: Net rate of production of reagent D as a function of the volumetric feed rate.

highest instantaneous selectivity. That is why the reactor is initially filled with A (to keep its concentration as high as possible) and B is added to it slowly (so it reacts nearly as fast as it is added and its concentration stays small).

$$S_{D/U,inst} = \frac{k_1}{k_2}\frac{C_A}{C_B} \tag{29}$$

Operating in that way introduces a trade-off between reaction rate and selectivity. A low concentration of B offers high selectivity, but it also results in both reaction rates being small. At the two ends of the spectrum, the rate at which D is produced is low. At low concentration of B it is produced very selectively, but at a very low rate. At high concentration of B, the rate is very high, but a large amount of U is being produced instead of D.

This can be seen by examination of the expression for the net rate of production of D, equation (30). Operation at low concentration of B produces a large amount of D, but requires a long processing time to do so. Operating at high concentration of B requires a short processing time, but also reduces the amount of D that is generated. Put differently, increasing the feed rate decreases both the numerator and the denominator in equation (30). At low feed rates the decrease in the denominator predominates and the net rate increases. At high feed rates, the numerator predominates and the rate decreases. The maximum net rate of production occurs at the feed rate where the change in the numerator and denominator are equal.

$$r_{D,net} = \frac{n_{D,f}}{t_f + t_{turn}} \tag{30}$$

Summary

The maximum net rate of production of reagent D is 7.51 x 10^{-4} lbmol h^{-1}, and it occurs when the volumetric feed rate is 11.1 gal h^{-1}.